

Bangabandhu Sheikh Mujibur Rahman University, Kishoreganj
B.Sc. in Computer Science and Engineering
Class Test 1, 1st Semester - 2023
Subject: CSE-1103, Discrete Mathematics

Time: 20 Minutes

Full Marks: 20

INSTRUCTIONS

- a. Answer all the questions
- b. Figures in the margin indicate **full marks**

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Question-1**10**

Consider two propositions p and q . Now explain the following terminologies.

- A. **Inverse** of $p \rightarrow q$
- B. **Contrapositive** of $p \rightarrow q$
- C. **Converse** of $p \rightarrow q$

Explain the validity of the following statement using a Truth Table.

“Although the **Inverse** and **Converse** of $p \rightarrow q$ are not equivalent, the **Contrapositive** is equivalent to $p \rightarrow q$.”

Question-2**04**

Illustrate the key difference between the *Disjunction* and *XOR* operations over two propositions with suitable examples.

Question-3**06**

Consider the following two propositions:

- p : Bangladesh wins if and only if Bangladesh plays at Mirpur.
- q : Either Bangladesh plays at Mirpur, or Bangladesh wins.

Prove using a truth table that both p and $\neg q$ have the same truth value.